Lattice Field Theory — Exam Questions

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Each chapter below has three questions of increasing difficulty: [Easy], [Intermediate], and [Hard].

1 Euclidean path integrals and lattice scalar field theory

Question 1. *[Easy]* *What is a Wick rotation, and why does rotating to Euclidean time make the path integral converge? Contrast the real-time weight $e^{iS/\hbar}$ with the Euclidean weight $e^{-S_E/\hbar}$.*

Question 2. *[Intermediate]* *Write down the discretized Euclidean action for a free scalar field on a hypercubic lattice of spacing a, and explain how a acts as an ultraviolet regulator via the Brillouin zone.*

Question 3. *[Hard]* *Derive the Euclidean path integral for a single non-relativistic particle starting from the transition amplitude $\langle x_f | e^{-iH(t_f - t_i)/\hbar} | x_i \rangle$. Show explicitly how the Trotter splitting of $H = p^2/2m + V(x)$ and insertion of complete sets of position and momentum eigenstates produces the discretized action, and state the continuum action recovered in the limit.*

2 Gauge theories on a lattice

Question 4. *[Easy]* *Define the link variable $U_\mu(n)$ and state its gauge transformation law. Why is a link variable, rather than the field A_μ directly, the natural object on the lattice?*

Question 5. *[Intermediate]* *Define the plaquette $U_{\mu\nu}(n)$ and write down the Wilson gauge action. Explain why the plaquette is the smallest gauge-invariant object built from links.*

Question 6. *[Hard]* *By expanding the link variables in powers of the lattice spacing a, show that the Wilson gauge action reduces to the continuum Yang–Mills action $\frac{1}{4}\int d^4x\, F^a_{\mu\nu} F^{a\,\mu\nu}$ as $a \to 0$. Identify the leading lattice-artifact correction.*

3 Fermions on a lattice

Question 7. *[Easy]* *State the fermion doubling problem. How many doublers does the naive discretization of the Dirac operator produce in d dimensions, and where in the Brillouin zone do they sit?*

Question 8. *[Intermediate]* *Compare Wilson and staggered fermions. What symmetry does each approach sacrifice or preserve, and by what mechanism does each remove (or reduce) the doublers?*

Question 9. *[Hard]* *State the Nielsen–Ninomiya no-go theorem and its assumptions. Explain how the Ginsparg–Wilson relation evades it and what form of lattice chiral symmetry it permits.*

4 Continuum limit

Question 10. [Easy] *What is the physical correlation length ξ_{phys} in terms of the lattice spacing a and the dimensionless correlation length ξ? Why must $\xi \to \infty$ (in lattice units) to reach the continuum?*

Question 11. [Intermediate] *Explain why the continuum limit must be taken at a second-order (critical) point of the lattice theory, and why a first-order transition is unsuitable.*

Question 12. [Hard] *Using asymptotic freedom and the two-loop beta function of $SU(N)$ gauge theory, derive how the lattice spacing a depends on the bare coupling g near the continuum limit. Explain what "asymptotic scaling" means and how it is used to set the lattice scale.*

5 Chemical potential and the numerical sign problem

Question 13. [Easy] *How is a quark chemical potential μ introduced into the lattice action, and why can it not simply multiply the number operator naively? State qualitatively why $\mu \neq 0$ leads to a complex action.*

Question 14. [Intermediate] *Explain the origin of the sign problem in terms of γ_5-Hermiticity of the Dirac operator. Why does reweighting from a positive ensemble fail as the volume grows?*

Question 15. [Hard] *Quantify the severity of the sign problem via the expectation value of the phase $\langle e^{i\theta} \rangle_{pq}$ in the phase-quenched ensemble, and show that it is suppressed exponentially in the volume, $\langle e^{i\theta} \rangle \sim e^{-V\Delta f}$. Relate Δf to a free-energy difference and explain the resulting cost of a reweighted Monte Carlo estimate.*

6 Techniques to circumvent the sign problem

Question 16. [Easy] *Distinguish sign quenching from phase quenching. Why is the phase-quenched theory generally not the theory one wants to solve?*

Question 17. [Intermediate] *Describe two conventional finite-density approaches (e.g. Taylor expansion in μ/T, imaginary chemical potential with analytic continuation, or reweighting). For each, state its domain of validity and its main limitation.*

Question 18. [Hard] *Explain the idea of Lefschetz thimbles: how deforming the integration contour into complexified field space onto steepest-descent manifolds tames the sign problem. What new difficulties (e.g. residual phase, multiple thimbles, Jacobian) does the method introduce?*